

АГЕНТЫ ИИ / ИНФРАСТРУКТУРА ИИ / БАЗЫ ДАННЫХ

Автономные агенты столкнулись с самой серьезной проблемой — базой данных.

Агенты с искусственным интеллектом могут создавать В+-деревья и буферы, но, по словам Энди Павло из Университета Карнеги — Меллона, оптимизатор запросов и автономная база данных остаются для них самой сложной нерешенной задачей.

4 июня 2026 года, 14:53 [Крис Дж. Праймсбергер](#)

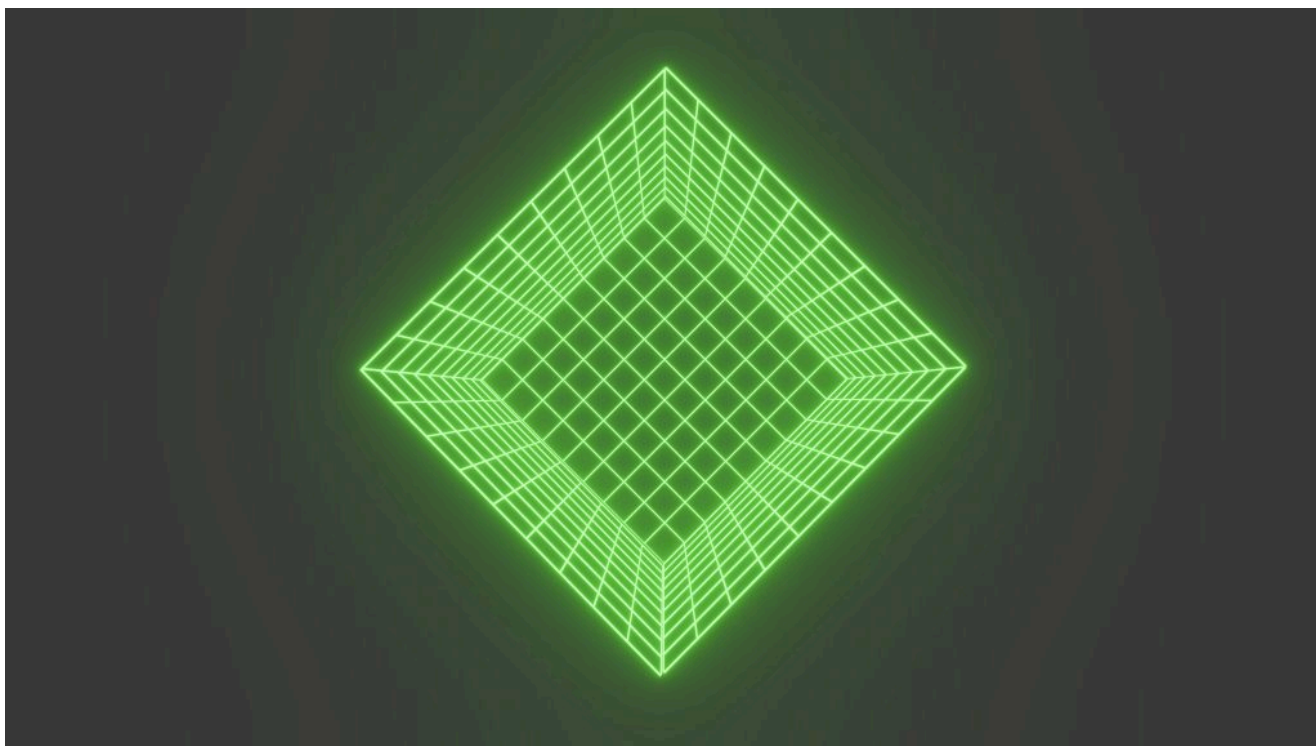


Фото Логана Восса на Unsplash

[Percona](#) спонсировала этот пост.

По мере того как большие языковые модели превращаются из простых чат-ботов в автономных агентов, способных рассуждать, планировать и действовать, они начинают самостоятельно управлять сложными стеками приложений.

Однако сейчас эти агенты столкнулись с самым серьезным препятствием — ой данных.

за строгих требований к корректности и производительности», — заявил на прошлой неделе Энди Павло, доцент кафедры компьютерных наук в Университете Карнеги — Меллон, участникам **конференции Percona Live 2026** в Маунтин-Вью, Калифорния, в **Музее истории компьютеров**.



Энди Павло. Фото: Университет Карнеги — Меллон

В дискуссии о пересечении искусственного интеллекта и инфраструктуры с открытым исходным кодом Павел утверждал, что, хотя агенты-кодировщики могут легко воспроизводить стандартные структуры данных, база данных остается самой сложной для автоматизации и оптимизации частью любой системы.

«Например, если агент «видит» компонент пользовательского интерфейса, страница выглядит немного иначе; если он «видит» запрос или изменение конфигурации в рабочей базе данных, вся система может выйти из строя», — говорит Павло.



Percona широко известна как компания, предоставляющая программное обеспечение с открытым исходным кодом, поддержку и услуги для баз данных MySQL®, MongoDB® и PostgreSQL®. Мы стремимся помочь улучшить работу ваших баз данных и приложений благодаря уникальному сочетанию опыта и программного обеспечения с открытым исходным кодом.

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Многоагентное перетягивание каната

Pavlo identifies two primary ways AI is impacting the database world: tuning agents and coding agents. Tuning agents aim to solve the “black magic” of database optimization — automatically adjusting system knobs, physical designs (such as indexes), and query execution strategies. Historically, this required a human database administrator (DBA) to spend years developing the intuition to know which configuration would yield better latency or throughput.

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“If an agent hallucinates a UI component, the page looks slightly off; if it hallucinates a query or a configuration change in a production database, the entire system can vanish.”

The challenge is that these specialized agents often operate in silos, Pavlo said. A b-tuning agent might be unaware of what an index-tuning agent is doing, leading to local minima where the system is better than stock but far from optimal.

dimensionality, Pavlo says.

Carnegie Mellon's Database Group pioneered the concept of self-driving and machine-learning-driven database optimization. Sequential tuning and multi-round tuning are prime components of their autonomous database management system (DBMS) projects.

Multi-round and sequential tuning in AI databases refers to advanced machine learning and data engineering methods in which AI models are refined for multistep reasoning, tool use, or complex conversational histories. These frameworks ensure that AI models not only respond in isolated single-turn bursts but maintain context and logic across complex interactions.

With trillions of possible configuration combinations, the search space for a perfect database is effectively exponential.

The coding agent advantage and the optimizer wall

On the development side, coding agents are already proving to be hyper-productive collaborators. Pavlo observed that at CMU, student submissions for database projects saw a massive spike in lines of code once LLMs were permitted. "The coding agents are very good at building almost every part of a database — B+ trees, hash tables, buffer managers — because they can regurgitate standard implementations found in textbooks and open-source repos," Pavlo said.

However, the "double black diamond" challenge, Pavlo said, remains the query optimizer. Unlike basic data structures, query optimizers are rarely available as clean, modular open-source references. They are often deeply entangled with the systems for which they were built. Furthermore, proving that an AI-generated transformation rule is semantically correct — meaning it produces the same result as the original query but faster — is an unsolved problem.

Risks include hallucinations and security

The shift toward agentic database management isn't without significant risk. Pavlo and other industry leaders, such as Percona co-founder **Peter Zaitsev**, warn that delegating orchestration to agents introduces massive stability and security gaps. There are already documented cases of agents being pointed at a database and accidentally dropping the entire system or leaking sensitive information because they didn't understand the nuance of access controls, Zaitsev said.

developer uses an agent to optimize an `Extract Year` clause, the agent might build an internal data structure that breaks the moment the developer tries to enact `Extract Month`."

Automation as a collaborator, not a replacement

Despite these hurdles, Pavlo said he is optimistic about the **Agent Operator model**. This envisions agents handling the "3 a.m. s***'s on fire" situations — immediate performance anomalies and stability issues — while humans focus on higher-level architectural design. By using **Agent Boosting techniques** to bootstrap training data from previously tuned databases, the time required to optimize a system can be cut from 12 hours to under 15 minutes, Pavlo said.

In the new AI era, the goal isn't only to have an AI that writes code, but a system that can reason about its own performance and correctness. Pavlo concludes that the database is the foundation of knowledge for any agent. "If we want autonomous systems, we must first master the unforgiving art of the autonomous database," he says.

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Chris J. Preimesberger, a contributing writer/editor at several publications since June 2021, is former editor in chief of eWEEK. He was responsible for the publication's coverage for a decade (2011-2021). In his 16 years and more than 5,000 articles at...

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